Churn Analysis and Predictive Modeling

Report

1. Introduction

This report provides an extensive analysis of student churn, utilising advanced machine learning techniques to predict and understand the factors leading to student drop-offs. The primary goal is to identify at-risk students early, allowing for timely and targeted interventions to improve retention rates. Understanding churn is critical in educational settings, as reducing drop-offs not only enhances student outcomes but also contributes to the sustainability and success of academic programs.

2. Methodology

The analysis was conducted in a structured and systematic manner, encompassing several key steps:

2.1 Data Cleaning

Data quality is fundamental to building effective predictive models. The cleaning process involved:

● **Handling Missing Values:** Critical fields such as age, gender, and duration metrics were checked for missing values. Rows with essential data missing were either imputed with appropriate values or removed to maintain data integrity.

● **Converting Data Types:** Date fields (e.g., "Opportunity Start Date," "Completion Date") were converted to the proper date-time format, enabling accurate time-based calculations and feature engineering.

2.2 Feature Engineering

To create meaningful features that improve model performance, the following transformations were performed:

● **Churn Label Creation:** A binary target variable, Churn, was defined. Students who had not completed their opportunities by the designated end date or had a missing completion date were labelled as churned (1), while others were labelled as non-churned (0).

● **Encoding Categorical Variables:** Variables such as Gender and City were encoded into numerical values using Label Encoding, making them suitable for input into machine learning algorithms.

● **Derived Features:** Calculated features such as "Time to Apply (days)" to measure the delay between opportunity availability and student application. These features aimed to capture the nuances of student engagement behaviour.

2.3 Model Building

Two machine learning models were developed and trained to predict churn:

Seven machine learning models were developed and trained to predict salary:

1. **Linear Regression**: A foundational model used for predicting continuous salary values, providing insights into relationships between features and salary.
2. **Random Forest Regressor**: An ensemble model that utilizes multiple decision trees to make robust and accurate predictions, particularly effective in handling non-linear relationships in the data.
3. **Support Vector Regressor (SVR)**: A model that uses support vector machines for regression tasks, effective in high-dimensional spaces.
4. **Gradient Boosting Regressor**: An ensemble model that builds trees sequentially, focusing on reducing errors made by previous models, leading to high predictive performance.
5. **XGBoost**: An optimized implementation of gradient boosting that offers speed and performance improvements.
6. **LightGBM**: A gradient boosting framework that uses tree-based learning algorithms and is designed for distributed and efficient training.
7. **K-Nearest Neighbors (KNN)**: A simple, instance-based learning method that predicts salary based on the average of the nearest data points.

2.4 Model Evaluation

Models were evaluated using a comprehensive set of metrics:

● **Accuracy:** Measures the overall correctness of predictions but can be misleading in

imbalanced datasets.

● **Precision:** Indicates the proportion of true positive predictions out of all positive predictions,

critical when false positives are costly.

● **Recall:** Reflects the model's ability to identify all actual churned cases, vital for minimizing

missed at-risk students.

● **F1 Score:** Balances precision and recall, providing a single metric for performance

comparison.

● **AUC (Area Under the Curve):** Measures the model's ability to distinguish between churned

and non-churned students, with higher values indicating better performance.

3. Results

3.1 Model Performance

Based on the provided results, we can analyze the performance of each model in terms of accuracy, AUC-ROC, precision, recall, and F1-score to determine which models are suitable for identifying students who are likely to drop out (class 1) and which ones are not. Here's a breakdown of the evaluation:

Performance Summary

1. **Logistic Regression**
   - Accuracy: **79.60%**
   - AUC-ROC: **0.796**
   - Precision for class 1 (Dropped Out): **0.00**
   - Recall for class 1: **0.00**
   - F1-Score for class 1: **0.00**
   - **Conclusion**: Poor performance in identifying dropouts. **Not suitable**.
2. **Decision Tree**
   - Accuracy: **92.01%**
   - AUC-ROC: **0.920**
   - Precision for class 1: **0.83**
   - Recall for class 1: **0.45**
   - F1-Score for class 1: **0.58**
   - **Conclusion**: Reasonable accuracy and ability to identify some dropouts, but recall is low. **Can be selected** with caution, further tuning may improve recall.
3. **Random Forest**
   - Accuracy: **95.59%**
   - AUC-ROC: **0.956**
   - Precision for class 1: **0.87**
   - Recall for class 1: **0.41**
   - F1-Score for class 1: **0.56**
   - **Conclusion**: High accuracy and good precision, but recall is still low. **Can be selected**.
4. **Gradient Boosting**
   - Accuracy: **95.33%**
   - AUC-ROC: **0.953**

- Precision for class 1: **0.87**
- Recall for class 1: **0.43**
- F1-Score for class 1: **0.57**
- **Conclusion**: Similar to Random Forest. **Can be selected**.

5. **Support Vector Machine**
   - Accuracy: **89.16%**
   - AUC-ROC: **0.892**
   - Precision for class 1: **0.68**
   - Recall for class 1: **0.36**
   - F1-Score for class 1: **0.47**
   - **Conclusion**: Lower accuracy and recall. **Not suitable**.
6. **K-Neighbors Classifier**
   - Accuracy: **85.59%**
   - AUC-ROC: **0.856**
   - Precision for class 1: **0.57**
   - Recall for class 1: **0.30**
   - F1-Score for class 1: **0.39**
   - **Conclusion**: Poor performance. **Not suitable**.
7. **XGBoost**
   - Accuracy: **95.46%**
   - AUC-ROC: **0.955**
   - Precision for class 1: **0.92**
   - Recall for class 1: **0.40**
   - F1-Score for class 1: **0.56**
   - **Conclusion**: High accuracy and precision, but recall is still low. **Can be selected**.

Recommendations

- **Select Models**: **Decision Tree**, **Random Forest**, **Gradient Boosting**, and **XGBoost** can be considered for further evaluation and possibly model tuning to improve recall for class 1 (Dropped Out).
- **Avoid Models**: **Logistic Regression**, **Support Vector Machine**, and **K-Neighbors Classifier** should be avoided as they exhibit poor performance, especially in identifying dropouts.

Final Thoughts

Identifying drop-off students is crucial, and given the context of your analysis, the **Decision Tree** could be a viable option to start with. However, it is recommended to further analyze and potentially tune the model to enhance its recall for the minority class. You may also consider ensemble methods or using class weighting to improve performance on the minority class in models like Random Forest and XGBoost.

3.2 Feature Importance

The feature importance analysis from various models highlights the key factors influencing student churn. Each model identifies distinct contributions of features to the overall predictions, providing valuable insights into the factors affecting student drop-off rates. Below is the breakdown of feature importance for the Decision Tree, Logistic Regression, Random Forest, Gradient Boosting, and XGBoost models.

Feature Importance for Decision Tree:

1. **Application Processing Time:** Contributed 39.0% to the model's predictions, indicating that the time taken to process applications significantly influences student retention.
2. **Program Duration:** Accounted for 27.0% of the importance, suggesting that the length of the program plays a crucial role in a student's decision to stay or drop out.
3. **Days to Start:** Contributed 10.0% to the importance, indicating that the timing of the program commencement affects student churn.
4. **Age:** Represented 5.0% of the importance, showing that age may have some impact on churn behavior.
5. **Days to End:** Also accounted for 5.0%, reflecting the relevance of remaining time in the program on retention decisions.
6. **Current/Intended Major (encoded):** Contributed 4.0%, indicating some influence of the major on student drop-off.
7. **Current Student Status (encoded):** Made up 3.0% of the importance, suggesting a minor impact on churn predictions.
8. **Opportunity Category (encoded):** Contributed 3.0%, showing some influence of the opportunity type on retention.
9. **Days to Graduate:** Accounted for 3.0%, indicating that proximity to graduation affects student retention.
10. **Country (encoded):** Had a minimal impact at 1.0%, suggesting that country of origin does not significantly influence churn predictions.

Feature Importance for Logistic Regression:

1. **Program Duration:** Contributed 62.0% to the model's predictions, highlighting its dominant role in determining student retention.
2. **Application Processing Time:** Accounted for 31.0% of the importance, indicating that faster processing may correlate with higher retention.
3. **Days to End:** Contributed 29.0%, suggesting that remaining time in the program significantly affects churn behavior.
4. **Age:** Had a negligible impact at 0.0%, indicating no significant influence on the prediction outcomes.
5. **Current/Intended Major (encoded):** Contributed 2.0%, showing a minor influence on churn decisions.

6. **Current Student Status (encoded):** Accounted for 7.0%, reflecting some relevance in the prediction model.
7. **Opportunity Category (encoded):** Contributed 1.36%, indicating a slight influence on retention.
8. **Days to Graduate:** Had a contribution of 0.9%, showing minimal impact.
9. **Country (encoded):** Contributed 0.4%, suggesting little influence.
10. **Gender (encoded):** Had a minimal impact at 0.5%, indicating gender does not play a significant role in churn predictions.

Feature Importance for Random Forest:

1. **Application Processing Time:** Contributed 27.0% to the model's predictions, indicating its significant role in student retention.
2. **Program Duration:** Accounted for 18.0%, highlighting the importance of program length on churn behavior.
3. **Days to Start:** Contributed 12.0%, indicating that timing influences retention rates.
4. **Days to End:** Accounted for 11.0%, reflecting how much time is left in the program can affect churn.
5. **Age:** Contributed 5.0%, showing some demographic influence on churn.
6. **Current/Intended Major (encoded):** Made up 4.0%, indicating minor importance.
7. **Opportunity Category (encoded):** Contributed 9.0%, suggesting some relevance in the model.
8. **Days to Graduate:** Had a contribution of 6.0%, indicating a slight influence on retention.
9. **Country (encoded):** Contributed 3.0%, suggesting minimal impact.
10. **Current Student Status (encoded):** Accounted for 3.0%, indicating a minor effect.

Feature Importance for Gradient Boosting:

1. **Application Processing Time:** Contributed 41.0%, suggesting a strong correlation with student retention.
2. **Program Duration:** Accounted for 28.0%, highlighting its importance in determining churn behavior.
3. **Days to End:** Contributed 8.0%, indicating its relevance in the model's predictions.
4. **Age:** Made up 6.0%, showing some demographic influence on churn.
5. **Days to Start:** Contributed 3.0%, indicating a minor role in retention decisions.
6. **Current/Intended Major (encoded):** Accounted for 2.0%, suggesting minimal impact.
7. **Current Student Status (encoded):** Contributed 1.0%, indicating a slight influence.
8. **Opportunity Category (encoded):** Had a contribution of 7.0%, showing some importance.
9. **Days to Graduate:** Accounted for 2.0%, indicating minor influence.
10. **Country (encoded):** Had a negligible impact at 0.0%.

Feature Importance for XGBoost:

1. **Application Processing Time:** Contributed 21.0%, showing its relevance in predicting student churn.
2. **Program Duration:** Accounted for 29.0%, indicating its significant impact on retention decisions.
3. **Opportunity Category (encoded):** Contributed 23.0%, highlighting its strong influence in churn predictions.
4. **Days to Start:** Made up 4.0%, suggesting a minor impact on retention.
5. **Days to End:** Contributed 8.0%, indicating some relevance in the model.
6. **Age:** Accounted for 2.0%, showing minimal impact.

7. **Current/Intended Major (encoded):** Contributed 2.0%, suggesting little influence on churn.
8. **Current Student Status (encoded):** Accounted for 7.0%, indicating minor relevance.
9. **Days to Graduate:** Contributed 1.0%, showing negligible impact.
10. **Gender (encoded):** Had a minimal impact at 1.0%, indicating gender does not play a significant role in churn predictions.

Conclusion:

The analysis of feature importance across various models indicates that **Program Duration** and **Application Processing Time** consistently emerge as critical factors influencing student retention. Other features, while relevant, show varying degrees of influence across models. The Decision Tree model, with its intuitive interpretability and strong performance, particularly highlights **Application Processing Time** and **Program Duration**, making it a suitable choice for understanding and predicting student churn effectively.

4. Recommendations

Based on the findings from the analysis, several strategies are recommended to improve student retention:

4.1 Optimize Opportunity Duration

● Since opportunity duration is the most critical factor, consider adjusting the length of opportunities to keep students engaged. Implementing milestones or periodic check-ins could help sustain interest and motivation over time.

4.2 Encourage Early Applications

● To address the significant impact of "Time to Apply," develop initiatives that prompt students to apply sooner. This could include personalized reminders, limited-time incentives, or streamlined application processes.

4.3 Tailor Support for Different Age Groups

● Age was identified as an influential factor. Institutions could provide age-specific support or resources, such as peer mentorship for younger students or flexible options for older students with additional responsibilities.

4.4 Implement Monitoring Systems

● Use the predictive model as part of a real-time monitoring system. This would allow administrators to flag and provide timely interventions for students at high risk of dropping off. Interventions could include one-on-one counselling, engagement activities, or academic support.

5. Conclusion

The analysis has successfully highlighted the key factors driving student churn and provided actionable insights for improving retention. By leveraging these findings, institutions can implement targeted strategies to support at-risk students, ultimately enhancing the educational experience and outcomes.

The use of predictive modelling, combined with data-driven interventions, holds the potential to significantly reduce churn and foster a more engaged and successful student body.